

Interpretation & Discussion

• What do the results of the confusion matrix indicate?

The confusion matrix reveals the model's ability to correctly classify malignant and benign tumors. High values in the diagonal cells (true positives and true negatives) indicate strong predictive performance. The low number of false positives and false negatives suggests the model rarely misclassifies cases, which is critical in medical diagnostics.

• How consistent is the model's performance based on 5-Fold Cross Validation?

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• What insights can be derived from the learning curve?

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• How can the model be improved?

To make our logistic regression model better, we can start by cleaning up the data and focusing only on the features that really matter. Some features don't help much and just add noise, so removing them makes the model faster and more accurate. We can also tweak the model's settings, like the regularization strength (called "C"), to help it generalize better and avoid overfitting. This means the model won't just memorize the training data—it will actually learn patterns that work well on new data too.

Besides improving logistic regression itself, we can try other models like Decision Trees, K-Nearest Neighbors (KNN), or Support Vector Machines (SVM) to see which one works best. Each model has its own strengths, and comparing them helps us choose the most reliable one. Another cool trick is using ensemble methods like bagging or boosting, which combine several models to make smarter predictions. These techniques can really boost accuracy and make the model more stable, especially when the data is tricky.